

Shweta Choudhary

Prime Minister Research Fellow



Personal Profile

Date of Birth 13 Nov 1999

Gender Female

Nationality Indian

Marital Single

Status

Address Department of Chemistry,
Indian Institute of Technology Roorkee,
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Area of Computational chemistry
research



Education

Dec 2022 **PhD in Computational Chemistry**, *Department of Chemistry, Indian Institute of Technology Roorkee, India*

2020–2022 **Master of Science in Chemistry**, *Department of Chemistry, Indian Institute of Technology Roorkee, India*, CGPA: 8.88/10

2017–2020 **Bachelor of Science with honor in Chemistry**, *Department of Chemistry, Gargi College, University of Delhi, India*, CGPA: 9.23/10

Research Experience

2021–2022 **Master's thesis**, *“First-principle investigation for anode material for beyond Li-ion batteries, focusing on designing low-dimensional anodes for Na-ion batteries”*, Indian Institute of Technology Roorkee, India

Advisor: **Prof. T. J. Dhilip Kumar**

Publications

- 2026 Dixit, D., **Choudhary, S.**, Sarkar, S., Banerjee, S. & Boving, T. Experimental and computational investigation of selective removal of ciprofloxacin by HKUST-1. *Journal of Water Process Engineering* **86**, 109802 (2026)[[DOI](#)]
- 2026 Goyal, N., **Choudhary, S.**, Witulski, B., Yan, S., Singh, B., Kornienko, N., Banerjee, S. & Mathur, S. Selective Dinitrogen Electroreduction to Ammonia on Zn₂V₂O₇ Pyrochlore. *ACS Catalysis* (2026)[[DOI](#)]
- 2025 **Choudhary, S.** & Banerjee, S. Ion coordination and migration mechanisms in alkali metal complex borohydride-based solid electrolytes. *Communications Chemistry* **8**, 123 (2025)[[DOI](#)]
- 2025 Saroha, R., **Choudhary, S.**, Brahma, R. & Banerjee, S. Designing superatomic (Li/Na/K)-Al-Sc cluster catalysts for selective electrochemical HER and NRR. *The Journal of Physical Chemistry C* **129**, 6562–6573 (2025)[[DOI](#)]
- 2024 **Choudhary, S.**, Saroha, R. & Banerjee, S. Efficient electron injection into graphullerene enables reversible NaC₂ sodium storage. *ACS Applied Materials & Interfaces* **16**, 50859–50869 (2024)[[DOI](#)]
- 2023 Saini, V.[†], Krishankant[†], **Choudhary, S.**[‡], Gaur, A., Banerjee, S., Bagchi, V. & Venkatesh, V. Atomically precise copper nanoclusters as a potential catalyst for the electrochemical oxygen evolution reaction. *Journal of Materials Chemistry A* **11**, 24754–24763 (2023) [‡]Second author[[DOI](#)]
- 2022 **Choudhary, S.**, Duhan, N. & Dhilip Kumar, T. J. Hydrogen passivated β_{12} -borophene nanoribbon: a propitious one-dimensional metallic anode for sodium-ion rechargeable batteries. *Applied Surface Science* **606**, 154825 (2022) [[DOI](#)]

Workshops, Presentations & Conferences

- Mar 2026 **Global Virtual Symposium: Computational Materials Discovery**, ACS Spring 2026. (Oral talk) [[Certificate](#)]
- Jan 2025 **Evolution of Electronic Structure Theory & Experimental Realization (EESTER)**, SRM Institute of Science and Technology & IIT Madras. (Attended workshop & Presented poster) [[Certificate](#)]
- Dec 2024 **Conference on Advances in Chemistry for Energy and Environment (CACEE)**, TIFR Mumbai. (Presented poster) [[Certificate](#)]
- Nov 2024 **Research Scholar's Day-ChemDay**, Department of Chemistry, IIT Roorkee. (Received best poster award) [[Certificate](#)]

Oct 2024 **CRSI-ACS Early Career Researchers' Symposium**, KIIT Bhubaneswar. (Received best poster award) [[Certificate](#)]

Jan 2024 **International Conference on Functional Materials**, IIT Kharagpur. (Presented poster) [[Certificate](#)]

———— Awards, Fellowships & Academic Achievements

2024 Best Poster Award, ChemDay 2024 at IIT Roorkee

2024 Best Poster Award, CRSI-ACS Early Career Researchers' Symposium 2024 at KIIT Bhubaneswar

2023 Prime Minister Research Fellowship (PMRF)

2022 Qualified Graduate Aptitude Test in Engineering (GATE)

2020 Qualified IIT- Joint Admission Test (JAM)

———— Experience with Theoretical Models, Methods & Packages

Models & methods Density functional theory (DFT), Harmonic and quasi-Harmonic approximation, density functional perturbation theory (DFPT), nudged elastic band (NEB) method, growing string method, classical and *ab initio* molecular dynamics simulations

Electronic structure codes VASP, FHI-aims, FHI-vibes, i-PI, Gaussian, GaussView, Quantum Espresso, Critic2, Material Studio, Avogadro, CP2K, and GROMACS

GUI & Python modules Pymatgen, ASE, VASPKIT code, baderVis, VTST tools, VESTA, VMD, sumo, pyband, Aimstools, GIMS, Phonopy, aimsChain

Technical skills Anaconda, Python, Bash/Shell scripting, LaTeX

Architectures Linux, Mac, Windows

———— Referees

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